NSN-T-001529b(DSA-DM)
June 22, 1970
Interim revision of
Fed. Spec. O-P-94a
September 13, 1955

INTERIM FEDERAL SPECIFICATION

TEST PAPER, HYDROGEN ION

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification O-P-94.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rolls and strips of indicator test papers suitable for determining the hydrogen ion concentration of solutions.

1.2 Classification.

1.2.1 Ranges, classes and styles. Indicator test papers shall be of the following ranges, classes, styles and unit of issue, as specified (See 6.2).

Wide range:

pH range 1 - pH 2.0 to 10.0 (even pH Units)
pH range 2 - pH 1.0 to 11.0 (odd pH Units)
pH range 3 - pH 1.0 to 11.0 (consecutive pH Units)
pH range 4 - pH 1.0 to 12.0

Medium range:

pH range 1 - pH 1.0 to 5.0
pH range 2 - pH 4.0 to 8.0
pH range 3 - pH 6.0 to 11.0
pH range 4 - pH 10.0 to 14.0

FSC 6630

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Short range:

pH range 1 - pH 6.0 to 8.0 (0.5 pH Units)
pH range 2 - pH 8.0 to 9.5 (0.5 pH Units)

Class:

A - Roll
B - Strips

Style:

1 - With Dispenser
2 - Without Dispenser

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Technical Association of Pulp and Paper Industry

TAPPI T 401m, Fiber Analysis of Paper and Paperboard
TAPPI T 410-0s, Basis Weight of Paper and Paperboard
TAPPI T 411m, Thickness and Density of Paper

(Application for copies should be addressed to Technical Association of the Pulp and Paper Industry, 155 East 44th Street, New York, New York

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3. REQUIREMENTS

3.1 Material. Indicator test apper shall be prepared from white, unsized filter paper which is composed of 100 - percent fully bleached fibers. The unimpregnated filter paper shall have a basis weight of 20 ≠ 2 pounds and shall have a thickness of 0.006 ≠ 0.001 inch (in). Tests shall be conducted in accordance with 4.4.2.

3.2 Classes.

3.2.1 Class A. Class A indicator paper shall be furnished in the form of rolls, 1/4 ≠ 1/6 in. wide by 15 feet ≠ 1, -0 in. long. The diameter of the roll shall be not more than 1-1/4 in.

3.2.2 Class B. Class B indicator paper shall be furnished in strips, 100 strips per vial, each strip 2 ≠ 1/8 in. long and 7/32 ≠ 1/32 in. wide.

3.3 Color changes.

3.3.1 Wide range. Wide range test papers shall show distinct and rapid color changes at the pH value on the comparison color chart (see 3.4) to the nearest pH unit when tested as specified in 4.4.1. Wide range test papers shall be furnished in the pH units as indicated by the appropriate pH range.

1 - pH range 2.0 to 10.0 (even pH units)
2 - pH range 1.0 to 11.0 (odd pH units)
3 - pH range 1.0 to 11.0 (consecutive pH units)
4 - pH range 1.0 to 12.0

3.3.2 Medium range. Medium range test papers shall show distinct and rapid color changes in steps of 1.0 pH units when tested as specified in 4.4.1. Medium range test papers shall be furnished in pH units as indicated by the appropriate pH range.

1 - pH 1.0 to 5.0
2 - pH 4.0 to 8.0
3 - pH 6.0 to 11.0
4 - pH 10.0 to 14.0

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3.3.3 Short range. Short range test papers shall show distinct and rapid color changes in steps of 0.5 pH units when tested as specified in 4.4.1. Short range test papers shall be furnished in pH units as indicated by the appropriate pH range.

1 pH 6.0 to 8.0 (0.5 pH units)
2 pH 8.0 to 9.5 (0.5 pH units)

3.4 Color chart. A color comparison chart shall be affixed to or accompany each dispenser, or if no dispenser is furnished, it shall accompany each package of paper or vial. The chart shall show the colors produced by the paper, and shall state the corresponding pH value required to produce each color.

3.5 Dispenser. Unless otherwise specified, dispenser shall be supplied with each roll, set of rolls, or pack of papers. Dispensers for roll paper shall be fabricated from transparent polystyrene or acrylic plastic material and shall be designed to hold two rolls (approximately 1-1/4 inches outside diameter and 1/4 inch wide. Roll dispensers shall be designed so that a convenient length can be torn for the test without opening the dispenser. The dispenser shall have a removable tightly fitting cover so that replacement rolls of test paper and color charts may be placed therein (see 6.2).

3.5.1 Replacement roll containers. Each replacement roll paper shall be wrapped in vinyl coated, heat-sealed aluminum foil, and then enclosed in amber colored polystyrene or acrylic plastic container. The cover shall fit tightly and securely over the container. The container shall be approximately 2 inches long by 2 inches high by 1-3/8 inches deep, and designed to hold five rolls.

3.6 Workmanship. Workmanship shall be in accordance with good commercial practice. The test paper shall be unfiro in color and size.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the filter paper, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. For Class A indicator paper, the unit of product for examination purposes shall be one roll, as specified, and for Class B indicator paper, the unit of product for examination shall be one vial, as specified.

Table I. Sampling For Examination

	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product for tests shall be as specified in 4.2. Acceptance number shall be zero.

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Table II. Sampling for tests

Component (Unit of Product)	Characteristic	Requirement	Test Procedure	Inspection level
Roll	Color change	3.3	4.4.1	S-2
Vial	Color change	3.3	4.4.1	S-2
Roll or Vial	Material identification	3.1	4.4.2.1	S-2
Roll or Vial	Basis weight	3.1	4.4.2.2	S-2
Roll or Vial	Thickness	3.1	4.4.2.3	S-2

4.3 Examination. The indicator paper shall be examined for defects including, but not limited to, those listed in Table III.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Table III

Classification of Defects

Categories	Defects
Major	
101	Color chart not furnished with dispenser, rolls, and strips.
102	Rolls, strips, and dispenser not properly identified as to pH range.
103	Dispenser cracked or deformed.
104	Rolls and strips not uniform in size and color.
105	Vials cracked or otherwise damaged.
Minor	
201	Rolls and strips not free from dirt, grease or other foreign material.
202	Unit package not properly marked with pH range and manufacturer's name or trademark.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Color reaction. To determine color reaction, the sample paper shall be tested using buffer solutions corresponding in pH to the values specified in 3.3.

4.4.2 Filter paper.

4.4.2.1 Material identification. Material identification shall be determined in accordance with TAPPI T 401m - 60.

4.4.2.2 Basis Weight. The basis weight of the pH paper shall be determined in accordance with TAPPI I 410-0s-61.

4.4.2.3 Thickness. The thickness of the pH paper shall be determined in accordance with TAPPI T 411m - 44.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified, (see 6.2).

5.1.1 Unit of issue. Two rolls of test paper, two color charts and one dispenser, as specified, constitutes one unit of issue: five rolls of test paper and one color chart, as specified, constitutes one unit of issue: or one hundred strips of test paper and one color chart, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit (optional)	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Immediate containers.

5.1.3.1.1 Class A, style 1. Two rolls of test paper and two color charts shall be supplied with a dispenser as specified in 3.5.

5.1.3.1.2 Class A, style 2. Five rolls of test paper and one color chart shall be packaged as specified in 3.5.1.

5.1.3.1.3 Class B. One hundred strips shall be packaged in a glass vial of appropriate size with a polyethylene stopper and color chart. Closure shall be adequate to prevent the entrance of moisture. Leakage test as specified in paragraph 4.4.3 of MIL-P-116 shall apply.

5.1.3.2 Unit package. At the option of the contractor, each unit shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be secured to prevent spilling of contents.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676, type 1, class A, style 4, except that commercial colors are acceptable. Box design shall include partitions fabricated of the same material used for fabrication of the box. Partitions shall form an individual snug-fitting cell for each immediate container. When unit boxes are used, partitions shall not be required. Closure shall be secured to prevent spilling of contents.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, grade V3C. Closure and waterproofing shall be as specified in the appendix of PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking of shipments shall be in accordance with MIL-STD-123.

5.3.2 Military Activities.

5.3.2.1 Immediate containers. Immediate containers shall be commercially identified.

5.3.2.2 Unit package. When furnished, each unit package shall be marked with item identification, stock number, pH range, level of packaging, date packed, quantity and unit. Marking shall include the legend: "SUBJECT TO DAMAGE BY LIGHT AND MOISTURE".

5.3.2.3 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. In addition, marking shall include the pH range and the legend: "SUBJECT TO DAMAGE BY LIGHT AND MOISTURE".

5.3.2.4 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend: "SUBJECT TO DAMAGE BY LIGHT AND MOISTURE".

6. NOTES

6.1 Intended use. Indicator test papers covered by this specification shall be used to determine the hydrogen-ion concentration of solutions.

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6.2 Ordering data. Procurement document should specify the following:

(a) Number and date of this specification
(b) FSN
(c) Specify range, class and style required (see 1.2.1)
(d) Specify if dispenser is not required (see 3.5)
(e) Selection of applicable levels of packing (see 5.1 and 5.2)

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 This specification covers the following items shown in the Federal Supply Catalog:

FSN	Item Identification
6630-442-8000	Test paper and color chart Hydrogen-ion and Dispenser;
6630-442-8010	Test paper and color chart, Hydrogen-ion
6630-442-8020	Test paper and color chart, Hydrogen-ion
6630-442-9005	Test paper and color chart, Hydrogen-ion, and dispenser
6630-442-9015	Test paper and color chart, Hydrogen-ion
6630-442-9025	Test paper and color chart, Hydrogen-ion

6.5 Classification changes. The following classification changes have been made:

New designation	Old designation O-P-94a
Wide range:	Long range:
pH range 1 -pH 2.0 to 10.0 (even pH Units)	-pH range a. pH 2 to 10
pH range 2 -pH 1.0 to 11.0 (odd pH Units)	-pH range b. pH 1 to 11
pH range 3 -pH 1.0 to 11.0 (consecutive pH Units)	-
pH range 4 -pH 1.0 to 12.0	-pH range c. pH 1 to 12

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New designation

Medium range:

pH range 1 - pH 1.0 to 5.0
pH range 2 - pH 4.0 to 8.0
pH range 3 - pH 6.0 to 11.0
pH range 4 - pH 10.0 to 14.0

Short range:

pH range 1 - pH 6.0 to 8.0 (0.5 pH Units)
pH range 2 - pH 8.0 to 9.5 (0.5 pH Units)

Class A Roll
Class B Strips

Style 1 With dispenser
Style 2 Without dispenser

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Old designation (cont.)
O-P-94a

Medium range:
pH range d. pH 1 to 5
pH range d. pH 1 to 5
pH range e. pH 4 to 8
pH range f. pH 6 to 11
pH range g. pH 10 to 14

Short range:

Class A. Ribbon
Class B. Strips

Preparing activity:
Defense Supply Agency - DM

Project Number: 6630-0156

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